



How to Derive $\sigma(t)$ in the Hull White Model.

★ Risk - Neutral Approach

⇒ The Hull White Model describes how interest rate 'r' moves over time. It has a dial called $\sigma(t)$.

Step 1. The model for the interest rate.

How the rates moves in the risk neutral world. Each tiny moment, rate gets a push from $\sigma(t)$, a pull back toward normal, and a random shock.

$$dr = (\sigma(t) - ar) dt + \sigma dW$$

∴

dr = The tiny change in the interest rate over a tiny amount of time.

$\sigma(t)$ = A push on the rate. = it can change over time, and it is the dial.

$-a \cdot r$ = A pull back when r is high, this term is a big negative number and drags the rate down. When r is low, it does the opposite. This is mean reversion. The bigger a is, the stronger the pull.

dt = a tiny step of time.

$\sigma \cdot dW$ ⇒ A random shock. σ is the size of the shock and dW is a random up or down kick.



$$dr = -a\left(\frac{Q(t)}{a} - r\right) dt + \sigma dw$$

Everything else is built from this equation. It is the rule that the rate follows.

Step 2 What a bond is worth (risk neutral pricing).

Ex A bond pays £1 at time T . Its price today is the average, Over all possible rate paths, of what £1 is worth after discounting at the moving rate r .

$$P(t, T) = E^Q \left[\exp \left(- \int_t^T r(s) ds \right) \mid F_t \right]$$

$P(t, T)$ = Today's price of the bond that pays £1 at time T .

$E^Q[\dots]$ = Take the average over all possible future rate paths in the risk neutral world. (Discounting factor)

$\exp(-\int r(s) ds)$ = The discount factor. The integral adds up all the interest earned from t to T . The minus sign and $\exp$ turn that total into how much £1 is worth today. (Average)



$1F-T \Rightarrow$ This means given everything we know today (time t)

helper function need later.

$$B(t, T) = \frac{1 - e^{-a(T-t)}}{a}$$

$B(t, T) \Rightarrow$ A measuring stick for how much the bond price react to the rate.

$1 - e^{-a(T-t)} \Rightarrow$ Grows from 0 to 1 as the time

gap $T-t$ gets larger.

$\div a \Rightarrow$ Dividing by a scales it. if a is tiny, B is about $T-t$, like an ordinary time to Maturity. if a is large, B is smaller, because the rate is pulled back quickly.

It gives us a bond price in terms of the rate path.

Once we can compute this average, we get the model bond price, and we can compare it with the real market price.



Step 3 Solve the rate equation.

To compute the average in step 2, we need $r(s)$ at every future time s . We solve the equation from step 1. Multiplying by the growth factor $\exp(a \cdot s)$ and adding up from t to s gives three parts.

$$r(s) = r(t)e^{-a(s-t)} + \int_t^s e^{-a(s-u)} Q(u) du + r \int_t^s e^{-a(s-u)} dW(u)$$

$r(t) \cdot e^{-a(s-t)}$ $\Rightarrow$ Today's rate, fading away as time passes, because the pull back drags it toward normal.

$\int_t^s e^{-a(s-u)} \cdot Q(u) \cdot du$ $\Rightarrow$ All the Q pushes collected from t up to s . A push from long ago has faded more than a recent one, and the $e^{-a(s-u)}$ factor does that fading.

$r \int_t^s e^{-a(s-u)} dW(u)$ $\Rightarrow$ All the random shocks collected from t up to s , also fading with age.

The rate $r(s)$ is the sum of three simple pieces: where it started, the pushes, and the shocks. Now we can add it up over time in step 4.



Step 4 : Average and Spread of the total interest

let X be the total interest accumulated from t to T . Because the shocks are bell-curved shaped, X is also bell curve shaped. A bell curve is fully described by two numbers: its average and its spread.

$$X \Rightarrow \int_t^T r(s) ds$$

$$E(X) = r(t) B(t, T) + \int_t^T \sigma(u) B(u, T) du$$

$E(X)$ $\Rightarrow$ Average total interest

$r(t) \cdot B(t, T) \Rightarrow$ The effect of the starting rate.

$\int \sigma(u) \cdot B(u, T) du \Rightarrow$ The effect of all the σ pushes.

Each push counts with weight $B(u, T)$.

$$\text{Var}(X) = \sigma^2 \int_t^T B(u, T)^2 du$$

$\text{Var}(X) \Rightarrow$ The Variance, which measures how spread out X is.

$\sigma^2 \Rightarrow$ The shock size, squared, bigger shocks mean more spread.

$\int B(u, T)^2 du \Rightarrow$ Add up how much each moment's shocks matters for the total.



Average of e^{-x} for a bell curve x there is a ready made rule.

$$E[e^{-x}] = \exp\left(-E[x] + \frac{\text{Var}[x]}{2}\right)$$

$\therefore E[e^{-x}] \Rightarrow$ The average of discount factor.

$-E[x] \Rightarrow$ The main part: discount by the average total interest.

$+ \text{Var}[x]/2 \Rightarrow$ A small upward bump caused by the spread. Because e^{-x} curves, randomness raises the average a little. This is called the Convexity effect.

The bond price in step 2 is exactly this average. So with the mean and spread in hand we can write the price in closed form.

Step 5

The bond price formula (with θ inside).

We put the average and spread into the rule from step 4 and take the natural log which turns the bond price into a sum

$$\ln P(T, t) = -r(t) B(t, T) - \int_t^T \theta(u) B(u, T) du + \frac{\sigma^2}{2} \int_t^T B(u, T)^2 du$$



$\ln P(t, T) \Rightarrow$ The log of model bond price:

$-r(t) \cdot B(t, T) \Rightarrow$ The effect of starting rate. A higher rate lowers the bond price

$-\int \theta(u) \cdot B(u, T) du \Rightarrow$ The effect of θ pushed. This is where our unknown θ appears.

$+(\sigma^2 / 2) \int (u, T)^2 du \Rightarrow$ The small convexity from randomness.

The model's bond prices depends on $\theta(t)$. If we choose $\theta(t)$ correctly, this price will equal the real mkt price.

Step 6 Match today's market Curve

Set $t=0$ meaning today. The mkt already gives us bond prices $P(0, T)$ for many maturities T . From those we can compute the forward rate $f(0, T)$ which is known data. We now require that the Model reproduces this $f(0, T)$

$$f(0, T) = \frac{-\partial \ln P(0, T)}{\partial T}$$

$f(0, T) \Rightarrow$ The forward rate for a tiny instant at time T , as implied by today's Market.

$-\partial \ln P(0, T) / \partial T \Rightarrow$ Take the Slope of log bond price With respect to Maturity T and flip



the sign. This is the definition of a forward rate.

Now differentiate the step 5 formula with respect to T .

Two facts make this easy: the slope of $B(u, T)$ with respect to T is $e^{-a(T-u)}$, and $B(T, T) = 1$. So the extra boundary terms disappear

$$f(0, T) = r(0)e^{-aT} + \int_0^T e^{-a(T-u)} Q(u) du - r^2 \int_0^T B(u, T) e^{-a(T-u)} du$$

$r(0) \cdot e^{-aT} \Rightarrow$ Today's rate, faded over time T .

$\int e^{-a(T-u)} \cdot Q(u) du \Rightarrow$ All the Q pushes from 0 to T , faded with age

$-r^2 \int B(u, T) \cdot e^{-a(T-u)} du \Rightarrow$ The Volatility term.

The Volatility integral can be worked out exactly. Substituting $t = T - u$ and integrating gives a neat answer.

$$\int_0^T B(u, T) e^{-a(T-u)} du = \frac{(1 - e^{-aT})^2}{2a^2}$$

To keep things tidy, give the Q integral its own name $g(T)$. It also satisfies a simple rule for its slope

$$g(T) = \int_0^T e^{-a(T-u)} Q(u) du, \quad g'(T) = Q(T) - ag(T)$$

$g(T) \Rightarrow$ Shorthand for the $\mathcal{O}$ integral above.

$g'(T) = \mathcal{O}(T) - a \cdot g(T) \Rightarrow$ The Slope of g . The newest $\mathcal{O}$ adds to it, and the old pushes and the old pushes fade at Speed a .

Putting it together, step 6 becomes.

$$f(0, T) = r(0) e^{-aT} + g(T) - \frac{\sigma^2}{2a^2} (1 - e^{-aT})^2$$

This is the matching Condition. The left side is the Mkt's forward rate. The right side is the model's because $\mathcal{O}$ sits inside $g(T)$, forcing the two sides to agree will pin $\mathcal{O}$ down

Step 7:

Get $\mathcal{O}$ by itself.

We want $\mathcal{O}$ alone. We differentiate the matching conditions once more with respect to T , so that the integral disappears and $\mathcal{O}(T)$ shows up directly.

$$\frac{\partial f}{\partial T} \Rightarrow -ar(0) e^{-aT} + \mathcal{O}(T) - ag(T) - \frac{\sigma^2}{a} (1 - e^{-aT})$$

$\partial f / \partial T \Rightarrow$ The Slope of Today's forward rate Curve. This is known from the market.

$-a \cdot r(0) \cdot e^{aT} \Rightarrow$ The slope of the first term

$Q(T) - a \cdot g(T) \Rightarrow$ The Slope of g , from the rule in step 6.

$-(r^2/a)(1 - e^{aT})e^{aT} \Rightarrow$ The Slope of the Volatility term.

The step above still has $g(T)$ in it. Use step 6 to replace it.

$$g(T) = f(0, T) - r(0) e^{-aT} + \frac{\sigma^2}{2a^2} (1 - e^{-aT})^2$$

When we substitute, the terms containing $r(0)$ cancel each other out. What is left is:

$$\frac{\partial f}{\partial T} = Q(T) - a f(0, T) - \frac{\sigma^2}{2a} (1 - e^{-aT})(1 + e^{-aT})$$

finally, tidy up the last product with a simple algebra fact, (With $x = e^{-aT}$, we have $(1-x)(1+x) = 1-x^2$)

$$(1 - e^{-aT}) / (1 + e^{-aT}) = 1 - e^{-2aT}$$

The Unknown integral has been eliminated. $Q(T)$ is now expressed using only the market's forward curve plus the model's a and σ .

Step 8The final Answer

$$Q(t) = \frac{\partial f(0, t)}{\partial t} + af(0, t) + \frac{\sigma^2}{2a} (1 - e^{-2at})$$

∴

$\partial f(0, t) / \partial t \Rightarrow$ The slope of today's forward rate Curve. If the mkt curve is rising the rate must drift upward along it.

$a \cdot f(0, t) \Rightarrow$ A correction for pull back. Since the Model drags the rate toward normal, I must add enough push to hold the rate on the mkt curve.

$(\sigma^2 / 2a) (1 - e^{-2at}) \Rightarrow$ The convexity correction from Volatility. It is small at first and grows with time.